Motion & Animation

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Easing & timing

Linear linear timing (constant speed) for continuous loops like spinners and tickers

Ease-out ease-out (decelerate) easing for things entering the screen, so they respond fast and settle gently

Ease-in ease-in (accelerate) easing on exits, so elements gather speed as they leave

Ease-in-out ease-in-out (S-curve) easing for moves between two resting positions on screen

CSS "ease" the default CSS ease curve, cubic-bezier(.25,.1,.25,1)

Steps stepped timing (steps) with no in-betweens, for a sprite-sheet or ticking-clock feel

Quart out a strong ease-out like easeOutQuart, cubic-bezier(.25,1,.5,1)

Expo out an expo ease-out, cubic-bezier(.16,1,.3,1): a fast start and a long, silky settle

Expo in an expo ease-in, cubic-bezier(.7,0,.84,0), so it barely moves and then launches away

Circ in-out a hard S-curve like easeInOutCirc, cubic-bezier(.85,0,.15,1), for decisive section changes

Back out a back-out easing that overshoots slightly and settles, cubic-bezier(.34,1.56,.64,1)

Back in a back-in easing that pulls back slightly before moving off, cubic-bezier(.36,0,.66,-.56)

Back in-out an anticipate-and-overshoot easing, cubic-bezier(.68,-.6,.32,1.6), on the hero element only

Material standard Material 3 standard easing, cubic-bezier(.2,0,0,1), for on-screen moves

Emphasized decelerate Material emphasized decelerate, cubic-bezier(.05,.7,.1,1), for important entrances; emphasized accelerate, cubic-bezier(.3,0,.8,.15), for their exits

Bouncy spring a bouncy spring (stiffness about 300, damping about 10) that overshoots and wobbles before settling

Critically damped spring a critically damped spring (damping ratio 1, no overshoot) so it arrives quickly and settles without bouncing

Gentle spring a gentle, low-stiffness spring (stiffness about 80) with one soft overshoot, calm and floaty

Snappy spring a stiff, snappy spring (stiffness about 500, damping about 28) with a barely visible overshoot

Bounce a bounce-out easing that lands and rebounds in two or three shrinking hops

Elastic an elastic-out easing that snaps past the target and vibrates into place

Inertia momentum after release: the element keeps the flick velocity and decays smoothly to a stop

Rubber banding rubber-band resistance when dragged past the edge, springing back on release

Micro duration micro durations of 100 to 150ms for small state changes like hovers, presses and toggles

Standard duration standard durations of 200 to 300ms for menus, tabs and panels

Long duration longer durations of 400 to 600ms only for full-screen transitions and hero moments

Duration scales with size scale duration with distance and size: about 150ms for small elements, up to 500ms for full-screen moves

Stagger stagger the list items 50ms apart as they fade and rise in

Overlap overlap the moves so each starts before the last finishes, flowing as one gesture

Choreography choreograph the reveal: the card scales in first, then the title, then the items stagger in

Enter slow, exit fast asymmetric timing: enter over 300ms with ease-out, exit over 200ms with ease-in

Hold add short holds (a beat of 300 to 500ms) between moves so each change registers

Reduced motion respect prefers-reduced-motion: replace slides, zooms and parallax with short crossfades

Principles of animation

Squash and stretch squash and stretch: the shape stretches when moving fast and squashes on impact, keeping its volume

Anticipation anticipation: a small crouch or pull-back in the opposite direction before the main move

Staging clear staging: only the key element moves while everything else holds still and dims

Pose to pose pose-to-pose animation with clear key poses and smooth in-betweens

Follow-through and overlap follow-through and overlapping action: loose parts lag behind the main body and settle after it stops

Slow in and slow out slow in and slow out: more frames near the start and end of each move, fewer in the middle

Arcs move along gentle arcs instead of straight lines, as living things do

Secondary action add a small secondary action, like an antenna bob or an icon wiggle, that supports the main move without stealing focus

Timing use timing to show weight: light things move quickly, heavy things take longer to start and stop

Exaggeration exaggerate the motion: a bigger squash, a longer wind-up and a higher arc than realism would allow

Solid drawing solid, volumetric forms that turn convincingly in space, with consistent perspective and light

Appeal give it appeal: simple readable shapes, a clear silhouette and small signs of life like blinking and breathing

Keyframes and tweening keyframe the key poses and let the tool interpolate between them with eased tweens

Overshoot and settle a small overshoot and settle as each panel lands

Smear frame use smear frames on the fastest moves: one stretched, streaky frame instead of motion blur

Motion blur natural directional motion blur on fast-moving elements, like a 180-degree film shutter

Moving hold a subtle moving hold: when at rest, keep a slow breathing scale and gentle drift so it never looks frozen

On twos animate on twos (12 drawings per second) for a choppy, hand-drawn feel

Weight make it feel heavy: slow to start, a hard landing and a small screen shake on impact

Cycle a seamless cycle that ends exactly where it starts, like a walk cycle, so it loops forever

Motion path animate along a curved motion path, auto-oriented so it turns to face the direction of travel

Shape morph morph the shape smoothly from one form to the next instead of swapping it

Transitions & cuts

Fade a simple fade-in (opacity 0 to 1 over 200ms) for content changes

Crossfade a fade-through: the old view fades out over 90ms, then the new one fades and scales in from 92 percent over 210ms

Push a push navigation transition: the new screen slides in from the right while the old one shifts left by a third with a slight dim

Cover present it as a cover: the new layer slides up from the bottom over the dimmed current screen

Reveal an uncover transition: the top layer slides away to reveal the screen already underneath

Zoom a zoom transition: the old view scales up to 110 percent and fades while the new one scales in from 90 percent

Shared element a shared-element transition: the tapped thumbnail grows and moves into the detail header while the rest crossfades

Container transform a container transform: the tapped card expands, morphing its shape and color, into the full detail screen

Card flip a 3D card flip around the vertical axis with perspective, revealing the back side

Circular reveal a circular reveal: the new view expands from the tap point as a growing circle (clip-path)

Expand and collapse expand the row in place, animating its height while the content below slides down smoothly

Directional tab slide switching tabs slides the content in the direction of the chosen tab while the indicator glides under it

Dialog pop the dialog scales in from 90 percent with a slight overshoot while a dark scrim fades in behind it

Hard cut hard cuts between shots, no transitions

Match cut a match cut: the sun at frame center cuts to the moon in exactly the same position and size

Jump cut jump cuts within the same framing to compress time, the subject popping to new positions

Smash cut a smash cut from the calm, quiet shot straight to a loud, chaotic one with no warning

J-cut and L-cut use J-cuts so the next scene's sound starts a beat before its picture

Dissolve a slow cross dissolve between the two shots to suggest time passing

Fade to black fade to black, hold a beat, then fade up on the next scene

Dip to white dip to white between the shots, like a flash of memory

Wipe a hard-edged horizontal wipe from left to right into the next shot

Iris an iris out: the frame closes to a shrinking circle around the subject, then to black

Whip-pan transition a whip-pan transition: both shots blur sideways through a fast swish

Hidden cut hide the cut under a dark foreground object passing the lens, so it reads as one continuous take

Zoom transition a zoom transition: punch hard into the first shot with zoom blur and land in the next shot mid-zoom

Flash and light leak cover the cut with a warm orange light leak, like light bleeding onto film

Glitch transition a short glitch transition with RGB split and horizontal slice jitter at the cut

Micro-interactions

Press buttons scale to 96 percent while pressed and spring back on release

Hover lift cards lift 4px with a larger, softer shadow on hover over 150ms

Ripple a Material-style ripple spreading from the touch point and fading out

Toggle switch the toggle knob slides with a snappy spring, stretching slightly mid-move, while the track fades to the accent color

Check draw when checked, the box fills and the checkmark draws itself in a quick stroke

Like burst on like, the heart pops to 120 percent with a spring, fills red and bursts a ring of small particles

Shake on an error, shake the field side to side three times over 400ms and show a red message below

Success check a success state: a circle draws around, then a checkmark ticks in with a small pop

Toast a toast slides up from the bottom with a spring, stays four seconds and slides back down

Label swap after copying, the button label slides up and out as "Copied ✓" slides in, reverting after 1.5 seconds

Tooltip tooltips appear after a 300ms hover delay, fading and sliding 4px toward the target, and vanish instantly on leave

Badge pop the notification badge pops in with an overshoot and bumps each time the count changes

Spinner a spinner whose arc grows and shrinks as it rotates, for short unknown waits

Typing dots a typing indicator of three dots bouncing in sequence inside a chat bubble

Skeleton shimmer skeleton placeholders in the shape of the real content with a soft shimmer sweeping left to right

Progress bar a determinate progress bar with a percent label, easing slightly faster toward the end

Indeterminate bar a thin indeterminate loading bar along the top edge, a segment sliding across repeatedly

Pull to refresh pull to refresh: the list drags down to reveal a spinner that fills with the pull, then springs back after loading

Number roll numbers roll up to their new value digit by digit, like an odometer, with an expo ease-out

Stateful button on submit, the button morphs into a circular spinner, then into a green checkmark when done

Drag and drop when dragged, the card lifts with a larger shadow and slight tilt, the drop zone highlights, and it springs into place on release

Reorder while reordering, the other rows slide smoothly out of the way to open a gap where the dragged row will land

Swipe to dismiss swipe to dismiss: the item follows the finger, fades past the threshold and flies off, then the list closes the gap

Swipe actions swiping a row left reveals Archive and Delete actions underneath, snapping open with a spring

Pinch to zoom pinch to zoom around the fingers' midpoint, easing to the nearest zoom level and rubber-banding past the limits

Bottom sheet detents a draggable bottom sheet with peek, half and full detents, snapping to the nearest one with a spring

Carousel snap a swipeable carousel that snaps each card to center, with the next card peeking and pager dots tracking

Long-press menu long-press: the item shrinks slightly while held, then lifts and a context menu blooms out from it

Scroll & cursor

Parallax subtle parallax: the background image scrolls at about 40 percent of the content's speed

Scroll reveal blocks fade in and rise 20px as they scroll into view, once only, with a 60ms stagger within each row

Pinned section pin the feature section while scrolling drives three steps inside it, then release

Scrollytelling scrollytelling: text steps scroll past while a sticky chart beside them updates at each step

Scroll progress a thin reading-progress bar at the top that fills as the page scrolls

Horizontal scroll section a pinned section where scrolling down moves a row of panels horizontally, then releases

Scroll-linked zoom as the page scrolls, the inset hero image expands to full-bleed, its corners squaring off

Scroll text highlight a large statement where each word fills from 20 percent to full opacity as the page scrolls

Shrinking header a large header that condenses into a slim sticky bar on scroll, the title shrinking into it

Marquee an infinite marquee band of large text scrolling left at a steady linear speed

Full-page snap full-screen sections that snap one at a time on scroll

Cursor follower a soft circular cursor follower that trails the pointer with lag and grows over links

Magnetic button a magnetic button that leans toward the nearby cursor and springs back when it leaves

3D hover tilt cards tilt up to 10 degrees in 3D toward the cursor, with a soft glare sliding across the surface

Hover spotlight a soft radial spotlight follows the cursor across dark cards, lighting up the borders it passes

Camera moves

Locked-off shot a locked-off static camera; only the clouds and grass move

Pan a slow pan right across the landscape from a fixed tripod

Tilt a slow tilt up from the ground to the sky

Zoom a slow optical zoom in on the figure, the camera itself not moving

Dolly in the camera dollies in toward the figure; the foreground slides out of frame while the mountains barely grow

Dolly out the camera dollies slowly back, leaving the figure small in the wide landscape

Truck the camera trucks left on a slider, with strong parallax between the foreground grass and the distant mountains

Pedestal the camera pedestals up slowly, rising vertically without tilting

Roll the camera slowly rolls into a 12-degree Dutch angle, the horizon tilting

Slow push-in a very slow, steady push-in over the whole shot, barely perceptible

Pull-back reveal start tight on the figure and pull back to reveal the vast empty landscape around them

Dolly zoom a dolly zoom: the figure stays the same size while the background swells and warps behind them

Orbit a slow orbit around the figure, keeping them centered as the background sweeps behind

Crane up a crane shot rising over the foreground to reveal the full landscape below

Tracking shot a tracking shot keeping pace beside the walking figure as the landscape streams past

Whip pan a whip pan: the camera snaps sideways so fast the frame blurs, landing on the next subject

Crash zoom a sudden crash zoom straight in on the figure

Rack focus a rack focus from the foreground grass to the distant mountains, shallow depth of field

Aerial flyover a smooth aerial drone flyover gliding forward and slightly down over the landscape at golden hour

FPV drone a fast FPV drone shot diving toward the figure and banking hard around them

Handheld a handheld camera with gentle, natural micro-shake, documentary style

Steadicam float a smooth floating Steadicam glide with gentle drift, no shake

POV shot a first-person POV shot walking forward, with a gentle head bob on each step

Bullet time bullet time: the moment freezes mid-action while the camera orbits around the figure

Time & video prompts

Slow motion in slow motion, like 120 frames per second played back at 24, every droplet and strand of hair visible

Timelapse a timelapse: clouds race across the sky and the light shifts from afternoon to dusk in seconds

Hyperlapse a hyperlapse rushing forward through the landscape, the light shifting fast

Speed ramp a speed ramp: real speed, then ramping down to slow motion at the peak of the jump, then snapping back to speed

Freeze frame the action freezes on a single frame mid-jump, slightly desaturated, while the music cuts out

Reverse motion reverse motion: scattered fragments fly back together into the whole object

Stop motion stop-motion animation at 12 frames per second with slight frame-to-frame jitter, tactile handmade materials

Cinemagraph a cinemagraph: the scene is a frozen photo except the river, which flows in a seamless loop

Seamless loop a seamless loop where the final frame matches the first exactly, so it repeats with no visible cut

Boomerang a boomerang loop: a one-second action plays forward and then backward, repeating

Shutter feel a narrow shutter for crisp, staccato motion with no motion blur, like the beach landing in Saving Private Ryan

One motion per shot one clear action per shot: the figure slowly turns toward the sun; nothing else changes

Camera and subject, separately Camera: slow push-in. Subject: the figure stands still, coat moving in the wind.

Ambient motion subtle ambient motion only: drifting dust in the light, grass swaying gently, slow-moving clouds

Say what stays still the figure, the horizon and the mountains stay completely still; only the sky and light change

Speed and duration words the camera moves very slowly and steadily over the full 8 seconds

Start and end frames use this image as the first frame and that one as the last; move smoothly between them with no extra action

Physical forces a steady warm wind blows from the left, pushing the grass and the figure's coat to the right

Keep identity keep the figure's face, clothing and proportions exactly consistent throughout; no morphing

Beat-by-beat timing 0–3s: static wide shot, dust drifting. 3–6s: slow push-in toward the figure. 6–8s: the figure turns toward the sun.

Motion strength low motion strength: subtle, gentle movement that preserves the original image

Motion styles

Kinetic typography kinetic typography: each word slams in on the beat, scaling and stacking in bold condensed type

Flat motion design flat 2D motion design: simple geometric shapes pop in with overshoot, morph and slide with snappy easing on a solid background

Line draw-on line art that draws itself on, stroke by stroke, as if by an invisible pen

Liquid and blobs gooey liquid blobs that merge and split organically, soft and colorful

2.5D parallax photo a 2.5D parallax effect: the still photo split into foreground, subject and background layers with a slow drifting camera

Isometric isometric motion graphics: blocks rise and assemble on a 30-degree grid with no perspective

Particles warm floating dust particles drifting slowly upward through the golden light

Data animation animated data: bars grow from the baseline with a stagger while the trend line draws left to right

Logo sting a three-second logo sting: shapes fly in and lock together into the mark on a sound hit, then the wordmark fades up

VHS and analog a VHS look: scan lines, chroma bleed, occasional tracking wobble and a camcorder date stamp

Hand-drawn 2D hand-drawn 2D cel animation on twos, flat colors with a gentle line boil

Rubber hose 1930s rubber-hose cartoon style: noodle limbs, pie-cut eyes, everything bouncing in rhythm, black and white film grain

Anime anime style: held dramatic poses, radial speed lines on the burst of action, painterly sky background

Rotoscope rotoscoped animation: realistic human motion traced over film, with flat color and drawn outlines

Claymation claymation: plasticine characters with visible fingerprints and slight frame-to-frame wobble, at 12 frames per second

Paper cut-out paper cut-out animation: flat textured paper shapes with soft drop shadows, moving in small jerky steps

Pixel art animation 16-bit pixel art animation: a small sprite with a 4-frame walk cycle on a crisp pixel grid, no smoothing

3D animated feature Pixar-style 3D animation: soft rounded forms, warm subsurface lighting and expressive squash-and-stretch

Comic-book painterly comic-book painterly animation like Spider-Verse: halftone shading, offset print colors, ink lines and characters animated on twos